

Software Test Plan (STP) – Online Quiz System

Project: Online Quiz System – console-based C++ MCQ quiz game

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the Online Quiz System v1.0. It sets out objectives, scope, strategy, resources, schedule, and responsibilities for testing, derived from the SRS v1.0 (06-09-2026).

Scope: Testing covers menu navigation and quiz setup, quiz gameplay, scoring and results, leaderboard/history, admin question-bank management, data persistence and error handling, non-functional requirements (performance, reliability, usability, maintainability, portability), and security requirements. GUI, web/mobile, networked/multiplayer play, and LMS integration are out of scope (per SRS §1.2).

References: Online Quiz System SRS v1.0; project README; Jira project OQSSMP (board 68); ISO/IEC/IEEE 29119 (test documentation).

Definitions: MCQ (Multiple Choice Question), CLI (Command Line Interface), SRS, RTM (Requirements Traceability Matrix), FR/NFR (Functional / Non-Functional Requirement), SR (Security Requirement), AC (Acceptance Criteria), UAT (User Acceptance Testing).

2. Test Items

- MenuModule – main menu, category and difficulty selection
- QuizEngine – question selection, shuffling, presentation, answer validation, skip/timeout
- ScoringModule – score calculation, results summary, grade/remark
- FileIOModule – question bank loading/saving, results persistence, leaderboard and history
- AdminModule – password authentication, logout, question CRUD and validation
- Build system – C++17 compilation on Windows, Linux, macOS

3. Features to be Tested

Features are mapped to SRS requirement IDs. Full test-case mapping is in Section 13.

Area	Requirements	Priority
Quiz setup	QZ-F-001 Main menu; QZ-F-002 Category selection; QZ-F-003 Difficulty selection; QZ-F-004 Random selection, no repeats	High / Medium
Quiz gameplay	QZ-F-010 Four-option MCQ display; QZ-F-011 Option shuffling; QZ-F-012 A–D input with re-prompt; QZ-F-013	High / Medium / Low

Area	Requirements	Priority
	Optional time limit; QZ-F-014 Skip; QZ-F-015 Immediate feedback	
Scoring & results	QZ-F-020 Score; QZ-F-021 Results summary; QZ-F-022 Grade/remark; QZ-F-023 Persist attempt; QZ-F-024 Leaderboard; QZ-F-025 Player history	High / Medium
Admin	QZ-F-030 Password-protected admin menu; QZ-F-031 Add; QZ-F-032 Edit/delete by ID; QZ-F-033 View/filter; QZ-F-034 Question validation	High / Medium
Data & errors	QZ-F-040 Load bank, clear error on missing/corrupt file; QZ-F-041 Immediate persistence; QZ-F-042 Invalid menu input handling	High
Non-functional	QZ-NF-001 Start-to-first-question ≤ 2 s; QZ-NF-002 No data loss on abrupt termination; QZ-NF-003 Usability; QZ-NF-004 Modularity; QZ-NF-005 Cross-platform build; QZ-NF-006 Data-driven categories	High / Medium / Low
Security	QZ-SR-001 Admin password required; SR-002 Answer hidden pre-submit; SR-003 Password not stored in plaintext; SR-004 File input validation; SR-005 No player-facing write path; SR-006 Lockout after 3 failed attempts	High / Medium

4. Features Not to be Tested

- GUI, web/mobile interfaces, networked or multiplayer play, LMS integration (out of scope per SRS)
- C++ standard library internals and compiler behaviour
- Operating-system terminal emulator behaviour beyond basic text I/O
- Concurrent multi-user access (system is single-user per session)
- Cloud/shared-file export of results (future extension, not in v1.0)

5. Test Approach / Strategy

Levels:

- Unit tests: module-level tests for scoring logic, answer/input validation, question validation, shuffling, and file parsing (Jira OQSSMP-25, OQSSMP-26).
- Integration tests: QuizEngine $\leftrightarrow$ ScoringModule $\leftrightarrow$ FileIOModule; AdminModule $\leftrightarrow$ FileIOModule (edits reflected in the next quiz).
- System tests: end-to-end console flows – start quiz $\rightarrow$ answer/skip $\rightarrow$ summary $\rightarrow$ saved result $\rightarrow$ leaderboard/history; admin login $\rightarrow$ CRUD.
- Acceptance tests (UAT): execution of the SRS acceptance suites: Menu Navigation, Quiz Gameplay, Scoring & Results, Admin Question Management, Data Persistence, Security, Usability.

Types:

- Functional testing (all QZ-F requirements)
- Regression testing (re-run smoke + affected suites after every fix and merge)
- Performance testing (startup-to-first-question time, per-question response)
- Reliability testing (forced termination, corrupt/missing files, invalid input)
- Usability testing (first-time user walkthrough without help)
- Portability testing (Windows, Linux, macOS builds)

- Boundary and negative testing (invalid menu choices, empty options, out-of-range values, oversized input)

Entry Criteria: Stable build compiles cleanly with `g++ -std=c++17`; question bank and results data files available; test environment ready; test cases reviewed.

Exit Criteria: 100% of planned test cases executed; all high-priority functional requirements implemented and verified; no critical NFR failures; 0 critical/high defects open; RTM shows all test cases passed (per SRS §6).

5.1 Security Validation

- Admin actions blocked without correct password (TC-Sec-01)
- Correct answer never shown on screen before submit/skip (TC-Sec-02)
- Admin password not recoverable from stored files or source – hashed (TC-Sec-03)
- Malformed question/results records logged and skipped without crash (TC-Sec-04)
- Code and UI review confirming no player-facing edit path for questions or scores (TC-Sec-05)
- 4th consecutive wrong password blocked until restart (TC-Sec-06)
- Fuzzing of console inputs (menu, answer, admin password, question fields) and of question/results files with malformed rows, delimiters, and oversized fields
- Manual tampering test on results file to confirm graceful handling (SO-2)

6. Test Environment

Hardware: Standard developer laptops (keyboard + monitor/terminal); no special peripherals.

Software: Compiled quiz executable (`g++ -std=c++17 -o quiz src/main.cpp`); Windows, Ubuntu Linux, and macOS terminals; sample question bank and results files from the data folder.

Tools: GoogleTest or Catch2 (unit tests), shell/PowerShell scripts with piped input (system tests), Valgrind / AddressSanitizer (memory errors), Git/GitHub (version control), Jira OQSSMP (defect and task tracking), spreadsheet for RTM (OQSSMP-27).

Test Data: Question bank with ≥ 3 categories $\times$ 3 difficulty levels (20–30 questions minimum, OQSSMP-15); edge-case banks (empty, missing, corrupt, duplicate IDs, empty option text, multiple/no correct answers); sample results file with 0, 1, and many records; test admin credentials.

7. Test Schedule

Dates are tentative; implementation had not started as of the repository status and will be adjusted with the sprint plan.

Milestones:

- Test plan review and approval: 02-Oct-2026
- Test case design and RTM setup: 03-Oct-2026 to 09-Oct-2026
- Environment setup and test data preparation: 07-Oct-2026
- Unit testing (alongside development): from 05-Oct-2026
- Integration and system test execution: 12-Oct-2026 to 23-Oct-2026
- Regression and retest: 24-Oct-2026 to 26-Oct-2026
- UAT: 27-Oct-2026 to 28-Oct-2026
- Test Summary Report: 29-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated) with IDs matching the SRS
- Unit test source files and test scripts
- Test Data (question banks, results files, edge-case files)
- Test Execution Logs
- Defect Reports (Jira)
- Updated RTM
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Ashvin Ronald Lobo	Prepare and maintain test plan and RTM, design and coordinate execution, verify acceptance criteria
Team Lead / Coordinator	Chiranth R.	Task allocation, build integration, release readiness, submission
Developer – Quiz Engine & Scoring	Abdul Mateen Shaikh	Unit tests for engine/scoring, fix defects in gameplay and scoring
Developer – Question Bank & Admin	Tanvi B. Shetty	Unit tests for file I/O, admin and validation, fix defects in those modules
Course Coordinator	Rajeshwari Adrakatti	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Implementation not started; compressed timeline for a 4-person team	Write test cases and unit tests in parallel with development; prioritize High requirements first
Delay in stable build delivery	Request early smoke builds; integrate weekly
Cross-platform differences (line endings, console input, timers) break portability or the optional time limit (QZ-F-013)	Test on all three OS targets early; keep time limit behind a feature flag
Access to macOS for testing	Use a teammate's machine or CI runner; document the fallback
Corrupt or hand-edited data files causing crashes	Dedicated malformed-file test data and fuzzing (QZ-SR-004, QZ-F-040)
SRS ambiguities (e.g., score bands, N in leaderboard, default admin password handling)	Clarify with team lead before test-case finalization; record decisions in the RTM comments

11. Assumptions & Dependencies

- Modules are delivered with the interfaces defined in the SRS (MenuModule, QuizEngine, ScoringModule, FileIOModule, AdminModule)
- Sample question bank and results files will be provided before execution
- A C++17 compiler is available on all three OS targets

- Score bands for grade/remark (QZ-F-022) will be defined before test-case finalization
- Jira board OQSSMP is the source of truth for task and defect status

12. Suspension & Resumption Criteria

Suspend testing if:

- Build fails to compile or crashes at startup
- Build is too unstable (blocks > 30% of test cases)
- Test environment or data files unavailable for > 4 hours

Resume testing if:

- Blocking defects are resolved and a new build passes smoke tests
- Environment and test data are restored

13. Test Case Management & Traceability

The RTM maps every SRS requirement to its test case. Status: N = Not started, P = Partial, A = Achieved/Passed. All start at N. Test cases are tracked in the RTM sheet (OQSSMP-27).

Req ID	Requirement	Priority	Test Case	Module	St.
QZ-F-001	Main menu with 4 options	High	TC-Menu-01	MenuModule	N
QZ-F-002	Category selection	High	TC-Menu-02	MenuModule	N
QZ-F-003	Difficulty selection	Medium	TC-Menu-03	MenuModule	N
QZ-F-004	Random selection, no duplicates	High	TC-Menu-04	QuizEngine	N
QZ-F-010	MCQ with options A–D	High	TC-Quiz-01	QuizEngine	N
QZ-F-011	Shuffle answer options	Medium	TC-Quiz-02	QuizEngine	N
QZ-F-012	A–D input, re-prompt on invalid	High	TC-Quiz-03	QuizEngine	N
QZ-F-013	Optional per-question time limit	Low	TC-Quiz-04	QuizEngine	N
QZ-F-014	Skip question	Medium	TC-Quiz-05	QuizEngine	N
QZ-F-015	Immediate feedback	Low	TC-Quiz-06	QuizEngine	N
QZ-F-020	Score calculation	High	TC-Score-01	ScoringModule	N
QZ-F-021	Results summary	High	TC-Score-02	ScoringModule	N
QZ-F-022	Grade/remark	Medium	TC-Score-03	ScoringModule	N
QZ-F-023	Persist attempt	High	TC-Score-04	FileIOModule	N
QZ-F-024	Leaderboard	Medium	TC-Score-05	FileIOModule	N
QZ-F-025	Player history	Medium	TC-Score-06	FileIOModule	N
QZ-F-030	Admin password protection	High	TC-Admin-01	AdminModule	N

Req ID	Requirement	Priority	Test Case	Module	St.
QZ-F-031	Add question	High	TC-Admin-02	AdminModule	N
QZ-F-032	Edit/delete by ID	High	TC-Admin-03	AdminModule	N
QZ-F-033	View/filter questions	Medium	TC-Admin-04	AdminModule	N
QZ-F-034	Validate question data	High	TC-Admin-05	AdminModule	N
QZ-F-040	Load bank; error on missing/corrupt	High	TC-Data-01	FileIOModule	N
QZ-F-041	Persist admin edits immediately	High	TC-Data-02	FileIOModule	N
QZ-F-042	Handle invalid menu input	High	TC-Data-03	MenuModule	N
QZ-NF-001	Startup ≤ 2 s	High	TC-Perf-01	FileIOModule / QuizEngine	N
QZ-NF-002	No data loss on termination	High	TC-Rel-01	FileIOModule	N
QZ-NF-003	Usability for first-time user	Medium	TC-UX-01	MenuModule / QuizEngine	N
QZ-NF-004	Modular code structure	Medium	TC-Maint-01	All	N
QZ-NF-005	Windows/Linux/macOS build	Medium	TC-Port-01	Build System	N
QZ-NF-006	Add categories via data only	Low	TC-Maint-02	FileIOModule	N
QZ-SR-001	Admin password required	High	TC-Sec-01	AdminModule	N
QZ-SR-002	Hide answer before submit	High	TC-Sec-02	QuizEngine	N
QZ-SR-003	Password stored hashed	Medium	TC-Sec-03	AdminModule	N
QZ-SR-004	Validate/sanitize file inputs	High	TC-Sec-04	FileIOModule	N
QZ-SR-005	No player-facing write path	High	TC-Sec-05	All	N
QZ-SR-006	Lockout after 3 failed attempts	Medium	TC-Sec-06	AdminModule	N

Key test scenarios (examples)

Test Case	Scenario / Expected result	Requirement
TC-Menu-04	Start 10-question quiz repeatedly; no question repeats within one attempt	QZ-F-004
TC-Quiz-03	Enter "E", "1", blank, "ab" at answer prompt; system re-prompts, score unchanged	QZ-F-012
TC-Score-02	Answer mix of correct/incorrect/skipped; correct + incorrect + skipped = total; percentage matches	QZ-F-021

Test Case	Scenario / Expected result	Requirement
TC-Admin-05	Add question with no correct answer, two correct answers, or empty option; rejected with clear error	QZ-F-034
TC-Data-01	Delete or corrupt questions file and launch; readable error, no crash	QZ-F-040
TC-Rel-01	Kill process mid-quiz and mid-admin-edit; previously saved data intact	QZ-NF-002
TC-Sec-06	Enter wrong admin password 3 times, then correct password; 4th attempt blocked until restart	QZ-SR-006

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed / blocked
- Defect density (defects per module) and defect severity distribution
- Defect aging
- Requirement coverage (requirements with at least one passed test case)
- Unit test code coverage for scoring, validation and file I/O modules

Reports:

- Daily/sprint execution status (via Jira and RTM)
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Ashvin Ronald Lobo	
Team Lead	Chiranth R.	
Course Coordinator	Rajeshwari Adrakatti	